

SOLIDWORKS API (.NET VBA C#) — Reference

(63/337): ICurveDrivenPatternFeatureData

SOLIDWORKS API Documentation (converted from CHM)

ICurveDrivenPatternFeatureData (Type)

ICurveDrivenPatternFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICurveDrivenPatternFeatureData Interface

Allows access to a curve-driven pattern feature.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ICurveDrivenPatternFeatureData
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData
```

C#

```
public interface ICurveDrivenPatternFeatureData
```

C++/CLI

```
public interface class ICurveDrivenPatternFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData](#).

Example

[Create and Access Curve-driven Pattern Feature \(C#\)](#)
[Create and Access Curve-driven Pattern Feature \(VB.NET\)](#)
[Create and Access Curve-driven Pattern Feature \(VBA\)](#)

Remarks

Read [Pattern Features and their Feature Data Objects](#).

Before calling [IFeatureManager::CreateDefinition](#), you must pre-select the entities needed to create the curve-driven pattern feature.

Use Mark=...	
To select...	
Features to pattern	4
Direction 1 entity (2D/3D curve, edge, sketch, or sketch entity)	1
Direction 2 entity (2D/3D curve, edge, sketch, or sketch entity)	2
Face normal (face on which the Direction 1 3D curve lies); needed only when ICurveDrivenPatternFeatureData::D1AlignmentMethod is set to swCurveDrivenPatternAlignment_e.swCurvePatternTangentToCurve and Direction 1 is a 3D curve entity	1024

After calling [IFeatureManager::CreateDefinition](#), you can explicitly set other properties on the feature data object.

You must call [IFeatureManager::CreateFeature](#) after initializing this feature data object in order to successfully create the curve-driven pattern feature.

For more information about curve-driven pattern features, see the **Parts and Features > Features > Patterns and Mirroring > Types of Patterns > Curve Driven Patterns and the Curve Driven Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

Accessors

[IFeature::GetDefinition](#)

[IFeatureManager::CreateDefinition](#)

Access Diagram

[CurveDrivenPatternFeatureData](#)

See Also

[ICurveDrivenPatternFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ILocalCurvePatternFeatureData Interface](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICurveDrivenPatternFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICurveDrivenPatternFeatureData Interface

For a list of all members of this type, see [ICurveDrivenPatternFeatureData members](#).

Public Methods

Name	Description
 AccessSelections	Gains access to the selections that define this curve-driven pattern feature.
 GetPatternBodyCount	Gets the number of seed bodies in this curve-driven pattern feature.
 GetPatternFaceCount	Gets a list of pattern faces for this curve-driven pattern feature.
 GetPatternFeatureCount	Gets the number of seed features for this curve-driven pattern feature.
 GetSkippedItemCount	Gets the number of skipped items for this curve-driven pattern feature.
 GetTransform	Gets the transform for the specified instance of this curve-driven pattern feature.
 IAccessSelections	Gains access to the selections that define this curve-driven pattern feature.
 IGetPatternBodyArray	Gets a list of the seed bodies for this curve-driven pattern feature.
 IGetPatternFaceArray	Gets a list of pattern faces for this curve-driven pattern feature.
 IGetPatternFeatureArray	Gets a list of pattern features in this curve-driven pattern feature.
 IGetSkippedItemArray	Gets the array of integers representing the skipped items for this curve-driven pattern feature.
 ISetPatternBodyArray	Sets the list of seed bodies for this curve-driven pattern feature.
 ISetPatternFaceArray	Sets a list of pattern faces for this curve-driven pattern feature.
 ISetPatternFeatureArray	Sets the list of pattern features for this curve-driven pattern feature.
 ISetSkippedItemArray	Sets the list of skipped items for this curve-driven pattern feature.
 ReleaseSelectionAccess	Releases access to the selections that define this curve-driven pattern feature.
 SetFeatureScope	Sets the feature scope, whether to autoselect the affected bodies, and the affected bodies in this curve-driven pattern feature.

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See Also

[ICurveDrivenPatternFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ILocalCurvePatternFeatureData Interface](#)

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ICurveDrivenPatternFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICurveDrivenPatternFeatureData Interface

The following tables list the members exposed by [ICurveDrivenPatternFeatureData](#).

Public Properties

Name	Description
 AutoSelect	Gets whether to automatically select all bodies in a multibody part intersected by this curve-driven pattern feature.

	BodyPattern	Gets or sets whether to base this curve-driven pattern feature on bodies or features and faces.
	D1AlignmentMethod	Gets or sets the alignment method for this curve-driven pattern feature in Direction 1.
	D1CurveMethod	Gets or sets the curve method for this curve-driven pattern feature in Direction 1.
	D1Direction	Gets or sets Direction 1 of this curve-driven pattern feature.
	D1FaceNormal	Gets or sets the face on which the 3D curve lies in Direction 1 of this curve driven pattern.
	D1InstanceCount	Gets or sets the number of instances in this curve-driven pattern in Direction 1.
	D1IsEqualSpaced	Gets or sets whether the pattern items are equally spaced in Direction 1.
	D1ReverseDirection	Gets or sets whether the pattern direction is reversed in Direction 1.
	D1Spacing	Gets or sets the spacing for this curve-driven pattern in Direction 1.
	D2Direction	Gets or sets Direction 2 of this bidirectional curve-driven pattern feature.
	D2InstanceCount	Gets or sets the number of instances in Direction 2 of this bidirectional curve-driven pattern feature.
	D2IsEqualSpaced	Gets or sets whether the pattern items are equally spaced in Direction 2 of this bidirectional curve-driven pattern feature.
	D2PatternSeedOnly	Gets or sets whether to replicate the seed pattern only in Direction 2, without replicating the curve pattern created under Direction 1.
	D2ReverseDirection	Gets or sets whether to reverse Direction 2 for this bidirectional curve-driven pattern feature.
	D2Spacing	Gets or sets the spacing of pattern instances in Direction 2 for this bidirectional curve-driven pattern feature.
	Dir2Specified	Gets or sets whether Direction 2 is specified for this curve-driven pattern feature.
	FeatureScope	Gets which bodies in this multibody part are affected by this curve-driven pattern feature.
	FeatureScopeBodies	Gets the bodies in this multibody part that are affected by this curve-driven pattern feature.
	GeometryPattern	Gets or sets whether to create the pattern using only the geometry (faces and edges) of the feature.
	PatternBodyArray	Gets or sets a list of the bodies to pattern for this curve-driven pattern.
	PatternElement	Gets or sets the type of entities to base this curve-driven pattern feature.
	PatternFaceArray	Gets or sets the list of faces to include in this curve-driven pattern feature.
	PatternFeatureArray	Gets or sets the list of features to include in this curve-driven feature pattern.
	PropagateVisualProperty	Gets or sets whether to propagate visual properties (e.g., colors, textures, etc.) to all curve-driven pattern instances.
	SkippedItemArray	Gets or sets the skipped items for this curve-driven pattern feature.
	VarySketch	Gets or sets whether to allow pattern instances of a seed sketch to vary in relation to enclosing geometry in this curve-driven pattern feature.

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Public Methods

Name	Description
 AccessSelections	Gains access to the selections that define this curve-driven pattern feature.
 GetPatternBodyCount	Gets the number of seed bodies in this curve-driven pattern feature.
 GetPatternFaceCount	Gets a list of pattern faces for this curve-driven pattern feature.
 GetPatternFeatureCount	Gets the number of seed features for this curve-driven pattern feature.
 GetSkippedItemCount	Gets the number of skipped items for this curve-driven pattern feature.
 GetTransform	Gets the transform for the specified instance of this curve-driven pattern feature.
 IAccessSelections	Gains access to the selections that define this curve-driven pattern feature.
 IGetPatternBodyArray	Gets a list of the seed bodies for this curve-driven pattern feature.
 IGetPatternFaceArray	Gets a list of pattern faces for this curve-driven pattern feature.
 IGetPatternFeatureArray	Gets a list of pattern features in this curve-driven pattern feature.
 IGetSkippedItemArray	Gets the array of integers representing the skipped items for this curve-driven pattern feature.
 ISetPatternBodyArray	Sets the list of seed bodies for this curve-driven pattern feature.
 ISetPatternFaceArray	Sets a list of pattern faces for this curve-driven pattern feature.

 ISetPatternFeatureArray	Sets the list of pattern features for this curve-driven pattern feature.
 ISetSkippedItemArray	Sets the list of skipped items for this curve-driven pattern feature.
 ReleaseSelectionAccess	Releases access to the selections that define this curve-driven pattern feature.
 SetFeatureScope	Sets the feature scope, whether to autoselect the affected bodies, and the affected bodies in this curve-driven pattern feature.

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See Also

[ICurveDrivenPatternFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ILocalCurvePatternFeatureData Interface](#)

SOLIDWORKS API Help

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ICurveDrivenPatternFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICurveDrivenPatternFeatureData Interface

For a list of all members of this type, see [ICurveDrivenPatternFeatureData members](#).

Public Properties

Name	Description
 AutoSelect	Gets whether to automatically select all bodies in a multibody part intersected by this curve-driven pattern feature.
 BodyPattern	Gets or sets whether to base this curve-driven pattern feature on bodies or features and faces.
 D1AlignmentMethod	Gets or sets the alignment method for this curve-driven pattern feature in Direction 1.
 D1CurveMethod	Gets or sets the curve method for this curve-driven pattern feature in Direction 1.
 D1Direction	Gets or sets Direction 1 of this curve-driven pattern feature.
 D1FaceNormal	Gets or sets the face on which the 3D curve lies in Direction 1 of this curve driven pattern.
 D1InstanceCount	Gets or sets the number of instances in this curve-driven pattern in Direction 1.
 D1IsEqualSpaced	Gets or sets whether the pattern items are equally spaced in Direction 1.
 D1ReverseDirection	Gets or sets whether the pattern direction is reversed in Direction 1.
 D1Spacing	Gets or sets the spacing for this curve-driven pattern in Direction 1.
 D2Direction	Gets or sets Direction 2 of this bidirectional curve-driven pattern feature.
 D2InstanceCount	Gets or sets the number of instances in Direction 2 of this bidirectional curve-driven pattern feature.
 D2IsEqualSpaced	Gets or sets whether the pattern items are equally spaced in Direction 2 of this bidirectional curve-driven pattern feature.
 D2PatternSeedOnly	Gets or sets whether to replicate the seed pattern only in Direction 2, without replicating the curve pattern created under Direction 1.
 D2ReverseDirection	Gets or sets whether to reverse Direction 2 for this bidirectional curve-driven pattern feature.
 D2Spacing	Gets or sets the spacing of pattern instances in Direction 2 for this bidirectional curve-driven pattern feature.
 Dir2Specified	Gets or sets whether Direction 2 is specified for this curve-driven pattern feature.
 FeatureScope	Gets which bodies in this multibody part are affected by this curve-driven pattern feature.
 FeatureScopeBodies	Gets the bodies in this multibody part that are affected by this curve-driven pattern feature.
 GeometryPattern	Gets or sets whether to create the pattern using only the geometry (faces and edges) of the feature.
 PatternBodyArray	Gets or sets a list of the bodies to pattern for this curve-driven pattern.
 PatternElement	Gets or sets the type of entities to base this curve-driven pattern feature.
 PatternFaceArray	Gets or sets the list of faces to include in this curve-driven pattern feature.
 PatternFeatureArray	Gets or sets the list of features to include in this curve-driven feature pattern.
 PropagateVisualProperty	Gets or sets whether to propagate visual properties (e.g., colors, textures, etc.) to all curve-driven pattern instances.
 SkippedItemArray	Gets or sets the skipped items for this curve-driven pattern feature.
 VarySketch	Gets or sets whether to allow pattern instances of a seed sketch to vary in relation to enclosing geometry in this curve-driven pattern feature.

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See Also

[ICurveDrivenPatternFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ILocalCurvePatternFeatureData Interface](#)

SOLIDWORKS API Help

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AccessSelections Method (ICurveDrivenPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : AccessSelections Method (ICurveDrivenPatternFeatureData)

Gains access to the selections that define this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function AccessSelections( _  
    ByVal TopDoc As System.Object, _  
    ByVal Component As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim TopDoc As System.Object  
Dim Component As System.Object  
Dim value As System.Boolean  
  
value = instance.AccessSelections(TopDoc, Component)
```

C#

```
System.bool AccessSelections(  
    System.object TopDoc,  
    System.object Component  
)
```

C++/CLI

```
System.bool AccessSelections(  
& System.Object^ TopDoc,  
& System.Object^ Component  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc

Top-level document

Component

Component in which the feature is to be modified

Return Value

True if the selections are successfully accessed, false if not

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::AccessSelections](#).

Example

[Create and Access Curve-driven Pattern Feature \(C#\)](#)
[Create and Access Curve-driven Pattern Feature \(VB.NET\)](#)
[Create and Access Curve-driven Pattern Feature \(VBA\)](#)

Remarks

To modify a feature in a...	Then...
Part	- TopDoc argument is the IModelDoc2 for the part - Component argument is NULL
Assembly	- TopDoc is the IModelDoc2 object for the assembly - Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) if you modified the feature
- [ICurveDrivenPatternFeatureData::ReleaseSelectionAccess](#) if you did not

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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AutoSelect Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : AutoSelect Property (ICurveDrivenPatternFeatureData)

Gets whether to automatically select all bodies in a multibody part intersected by this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property AutoSelect As System.Boolean

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Boolean
```

```
value = instance.AutoSelect
```

C#

```
System.bool AutoSelect {get;}
```

C++/CLI

```
property System.bool AutoSelect {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to automatically select all bodies intersected by this curve driven pattern feature, false to specify affected bodies

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::AutoSelect](#).

Remarks

This property is valid only if [ICurveDrivenPatternFeatureData::GeometryPattern](#) is true.

Call [ICurveDrivenPatternFeatureData::SetFeatureScope](#) to set this property.

For more information, see the **Curve Driven Patterns and the Curve Driven Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

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BodyPattern Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : BodyPattern Property (ICurveDrivenPatternFeatureData)

Gets or sets whether to base this curve-driven pattern feature on bodies or features and faces.

.NET Syntax

Visual Basic (Declaration)

Property BodyPattern As System.Boolean

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Boolean
```

```
instance.BodyPattern = value
```

```
value = instance.BodyPattern
```

C#

```
System.bool BodyPattern {get; set;}
```

C++/CLI

```
property System.bool BodyPattern {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True for bodies, false for features and faces

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::BodyPattern](#).

Remarks

If this property is:

- True, use [ICurveDrivenPatternFeatureData::PatternBodyArray](#) to specify the seed bodies to pattern.
- False, use [ICurveDrivenPatternFeatureData::PatternFaceArray](#) or [ICurveDrivenPatternFeatureData::PatternFeatureArray](#) to specify the face or feature seeds to pattern.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::PatternElement Property \(\)](#)
[ICurveDrivenPatternFeatureData::IGetPatternBodyArray Method \(\)](#)
[ICurveDrivenPatternFeatureData::IGetPatternFaceArray Method \(\)](#)
[ICurveDrivenPatternFeatureData::IGetPatternFeatureArray Method \(\)](#)
[ICurveDrivenPatternFeatureData::ISetPatternBodyArray Method \(\)](#)
[ICurveDrivenPatternFeatureData::ISetPatternFaceArray Method \(\)](#)
[ICurveDrivenPatternFeatureData::ISetPatternFeatureArray Method \(\)](#)

Availability

SOLIDWORKS 2015 SP4, Revision Number 23.4

SOLIDWORKS API Help

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D1AlignmentMethod Property (ICurveDrivenPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : D1AlignmentMethod Property (ICurveDrivenPatternFeatureData)

Gets or sets the alignment method for this curve-driven pattern feature in Direction 1.

.NET Syntax

Visual Basic (Declaration)

Property D1AlignmentMethod As System.Integer

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Integer
```

```
instance.D1AlignmentMethod = value
```

```
value = instance.D1AlignmentMethod
```

C#

```
System.int D1AlignmentMethod {get; set;}
```

C++/CLI

```
property System.int D1AlignmentMethod {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Alignment method as defined in [swCurveDrivenPatternAlignment_e](#)

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::D1AlignmentMethod](#).

Example

[Create and Access Curve-driven Pattern Feature \(C#\)](#)
[Create and Access Curve-driven Pattern Feature \(VB.NET\)](#)
[Create and Access Curve-driven Pattern Feature \(VBA\)](#)

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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D1CurveMethod Property (ICurveDrivenPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : D1CurveMethod Property (ICurveDrivenPatternFeatureData)

Gets or sets the curve method for this curve-driven pattern feature in Direction 1.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property D1CurveMethod As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Integer  
  
instance.D1CurveMethod = value  
value = instance.D1CurveMethod
```

C#

```
System.int D1CurveMethod {get; set;}
```

C++/CLI

```
property System.int D1CurveMethod {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Curve method as defined in [swCurveDrivenPatternCurveMethod_e](#)

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::D1CurveMethod](#).

Example

[Create and Access Curve-driven Pattern Feature \(C#\)](#)
[Create and Access Curve-driven Pattern Feature \(VB.NET\)](#)
[Create and Access Curve-driven Pattern Feature \(VBA\)](#)

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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D1Direction Property (ICurveDrivenPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : D1Direction Property (ICurveDrivenPatternFeatureData)

Gets or sets Direction 1 of this curve-driven pattern feature.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property D1Direction As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Object
```

```
instance.D1Direction = value  
value = instance.D1Direction
```

C#

```
System.Object D1Direction {get; set;}
```

C++/CLI

```
property System.Object^ D1Direction {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pattern direction entity (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::D1Direction](#).

Example

[Create and Access Curve-driven Pattern Feature \(C#\)](#)
[Create and Access Curve-driven Pattern Feature \(VB.NET\)](#)
[Create and Access Curve-driven Pattern Feature \(VBA\)](#)

Remarks

The pattern direction can be a curve, edge, sketch entity, or a sketch. You must pre-select the direction entity using selection Mark = 1 before creating the feature definition. If you are using a 3D curve for Direction 1, then you must also pre-select the face normal entity using selection Mark = 1024.

Use this property only when editing the pattern feature.

If you specify this property using a 3D curve, you should also specify [ICurveDrivenPatternFeatureData::D1FaceNormal](#).

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::D2Direction Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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D1FaceNormal Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : D1FaceNormal Property (ICurveDrivenPatternFeatureData)

Gets or sets the face on which the 3D curve lies in Direction 1 of this curve driven pattern.

.NET Syntax

Visual Basic (Declaration)

```
Property D1FaceNormal As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Object
```

```
instance.D1FaceNormal = value
```

```
value = instance.D1FaceNormal
```

C#

```
System.Object D1FaceNormal {get; set;}
```

C++/CLI

```
property System.Object^ D1FaceNormal {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[IFace2](#)

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::D1FaceNormal](#).

Example

```
' VBA
```

```
,
```

```
' This example shows how to pre-select the face normal entity for creating a curve-driven pattern  
' that uses a helical curve for Direction 1.
```

```
,
```

```
' Preconditions: Ensure that the specified part template exists.
```

```
,
```

```
' Postconditions: Inspect the graphics area and the FeatureManager design tree
```

```
'=====
```

```
Dim swApp As SldWorks.SldWorks  
Dim swFeat As SldWorks.Feature  
Dim swFeatMgr As SldWorks.FeatureManager  
Dim swFeatData As SldWorks.CurveDrivenPatternFeatureData  
Dim swPart As SldWorks.PartDoc  
Dim Part As SldWorks.ModelDoc2  
Dim skSegment As SldWorks.SketchSegment  
Dim myFeature As SldWorks.Feature  
Dim vSkLines As Variant  
Dim swSheetWidth As Double  
Dim swSheetHeight As Double  
Dim boolstatus As Boolean  
Dim longstatus As Long, longwarnings As Long
```

```
Option Explicit
```

```
Sub main()
```

```
    Set swApp = Application.SldWorks
```

```
    'Create new part that includes 3 features: Boss-Extrude1, Boss-Extrude2, and Helix/Spiral1
```

```
    swSheetWidth = 0
```

```
    swSheetHeight = 0
```

```
    Set Part = swApp.NewDocument("C:\ProgramData\SOLIDWORKS\SOLIDWORKS 2021\templates\Part.PRTDOT", 0, swSheetWidth,  
swSheetHeight)
```

```
    Set swPart = Part
```

```
    swApp.ActivateDoc2 "Part1", False, longstatus
```

```
    Set Part = swApp.ActiveDoc
```

```
    Part.SketchManager.InsertSketch True
```

```
    Set skSegment = Part.SketchManager.CreateCircle(0#, 0#, 0#, 0.31751, 0#, 0#)
```

```
    Part.ClearSelection2 True
```

```
    Part.SketchManager.InsertSketch True
```

```
    Set myFeature = Part.FeatureManager.FeatureExtrusion2(True, False, False, 0, 0, 1.029208, 0.00254, False, False, False, False,  
1.74532925199433E-02, 1.74532925199433E-02, False, False, False, False, True, True, True, 0, 0, False)
```

```
    Part.SelectionManager.EnableContourSelection = False
```

```
    Part.SketchManager.InsertSketch True
```

```
    boolstatus = Part.Extension.SelectByRay(-8.19439014833279E-03, -2.48002522846491E-03, 8.636000000000702E-02, -0.400036026779312  
-0.515038074910024, -0.758094294050284, 8.9519512195122E-04, 2, False, 0, 0)
```

Part.ClearSelection2 True

```
vSkLines = Part.SketchManager.CreatePolygon(-0.19, -0.01, 0, -0.19, 0.05, 0, 6, True)
Set myFeature = Part.FeatureManager.FeatureExtrusion2(True, False, True, 0, 0, 0.24638, 0.00254, False, False, False, False,
1.74532925199433E-02, 1.74532925199433E-02, False, False, False, False, True, True, True, 0, 0, False)
Part.ClearSelection2 True
```

```
boolstatus = Part.Extension.SelectByID2("Front Plane", "PLANE", 0, 0, 0, False, 0, Nothing, 0)
Set skSegment = Part.SketchManager.CreateCircle(0#, 0#, 0#, 0.3175, 0, 0)
Part.InsertHelix False, True, False, False, swHelixDefinedBy_e.swHelixDefinedByPitchAndRevolution, 1.016, 1.0167, 1, 0, 3.926
```

```
'Pre-select Helix/Spiral1 for Direction 1 entity
boolstatus = Part.Extension.SelectByID2("Helix/Spiral1", "REFERENCECURVES", 0, 0, 0, False, 1, Nothing, 0)
'Pre-select Boss-Extrude2 as feature to pattern
boolstatus = Part.Extension.SelectByID2("Boss-Extrude2", "BODYFEATURE", 0, 0, 0, True, 4, Nothing, 0)
'Pre-select face on Boss-Extrude1 as face normal entity
boolstatus = Part.Extension.SelectByRay(0.298048689727977, -0.109451261534383, 0.233585198055067, -0.903801459854838,
2.28031013589574E-02, 0.427344053114907, 7.98164962810585E-03, 2, True, 1024, 0)
```

Set swFeatMgr = Part.FeatureManager

```
'Create curve driven pattern feature data object
Set swFeatData = swFeatMgr.CreateDefinition(swFeatureNameID_e.swFmCurvePattern)
```

```
swFeatData.D1AlignmentMethod = 0
swFeatData.D1CurveMethod = 0
swFeatData.D1InstanceCount = 8
swFeatData.D1IsEqualSpaced = True
swFeatData.D1ReverseDirection = False
swFeatData.D1Spacing = 0.00254
swFeatData.D2InstanceCount = 1
swFeatData.D2IsEqualSpaced = False
swFeatData.D2PatternSeedOnly = False
swFeatData.D2ReverseDirection = False
swFeatData.D2Spacing = 0.00254
swFeatData.Dir2Specified = False
```

```
'Create CrvPattern1 feature
Set swFeat = swFeatMgr.CreateFeature(swFeatData)
```

Part.ViewZoomtofit2

End Sub

Remarks

This property is valid only when:

- editing the curve-driven pattern feature
- ICurveDrivenPatternFeatureData::D1AlignmentMethod is set to swCurveDrivenPatternAlignment_e.swCurvePatternTangentToCurve
- specifying Direction 1 using a 3D curve

To create a curve-driven pattern feature that uses a 3D curve for Direction 1, you must pre-select the face normal with selection Mark = 1024 before creating the curve-driven pattern feature data object.

If you try to edit this property to null or Nothing, [IFeature::ModifyDefinition](#) returns true but the property does not change.

For more information, see the **Curve Driven Patterns and the Curve Driven Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1InstanceCount Property (ICurveDrivenPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : D1InstanceCount Property (ICurveDrivenPatternFeatureData)

Gets or sets the number of instances in this curve-driven pattern in Direction 1.

.NET Syntax

Property D1InstanceCount As System.Integer



```
Dim instance As ICurveDrivenPatternFeatureData
Dim value As System.Integer

instance.D1InstanceCount = value

value = instance.D1InstanceCount
```

```
System.int D1InstanceCount {get; set;}
```

```
property System.int D1InstanceCount {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Number of instances

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::D1InstanceCount](#).

Example

[Create and Access Curve-driven Pattern Feature \(C#\)](#)
[Create and Access Curve-driven Pattern Feature \(VB.NET\)](#)
[Create and Access Curve-driven Pattern Feature \(VBA\)](#)

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::D2InstanceCount Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

D1IsEqualSpaced Property (ICurveDrivenPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : D1IsEqualSpaced Property (ICurveDrivenPatternFeatureData)

Gets or sets whether the pattern items are equally spaced in Direction 1.

.NET Syntax

Property D1IsEqualSpaced As System.Boolean



```
Dim instance As ICurveDrivenPatternFeatureData
Dim value As System.Boolean

instance.D1IsEqualSpaced = value
```

```
value = instance.D1IsEqualSpaced
```

C#

```
System.bool D1IsEqualSpaced {get; set;}
```

C++/CLI

```
property System.bool D1IsEqualSpaced {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True spaces the pattern items equally, false does not

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::D1IsEqualSpaced](#).

Example

[Create and Access Curve-driven Pattern Feature \(C#\)](#)
[Create and Access Curve-driven Pattern Feature \(VB.NET\)](#)
[Create and Access Curve-driven Pattern Feature \(VBA\)](#)

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::D2IsEqualSpaced Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1ReverseDirection Property (ICurveDrivenPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : D1ReverseDirection Property (ICurveDrivenPatternFeatureData)

Gets or sets whether the pattern direction is reversed in Direction 1.

.NET Syntax

Visual Basic (Declaration)

```
Property D1ReverseDirection As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Boolean
```

```
instance.D1ReverseDirection = value
```

```
value = instance.D1ReverseDirection
```

C#

```
System.bool D1ReverseDirection {get; set;}
```

C++/CLI

```
property System.bool D1ReverseDirection {  
    System.bool get();
```

```
void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True reverses the pattern direction, false does not

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::D1ReverseDirection](#).

Example

[Create and Access Curve-driven Pattern Feature \(C#\)](#)
[Create and Access Curve-driven Pattern Feature \(VB.NET\)](#)
[Create and Access Curve-driven Pattern Feature \(VBA\)](#)

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::D2ReverseDirection Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1Spacing Property (ICurveDrivenPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : D1Spacing Property (ICurveDrivenPatternFeatureData)

Gets or sets the spacing for this curve-driven pattern in Direction 1.

.NET Syntax

Visual Basic (Declaration)

Property D1Spacing As System.Double

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Double
```

```
instance.D1Spacing = value
```

```
value = instance.D1Spacing
```

C#

```
System.double D1Spacing {get; set;}
```

C++/CLI

```
property System.double D1Spacing {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pattern spacing

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::D1Spacing](#).

Example

[Create and Access Curve-driven Pattern Feature \(C#\)](#)
[Create and Access Curve-driven Pattern Feature \(VB.NET\)](#)
[Create and Access Curve-driven Pattern Feature \(VBA\)](#)

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::D2Spacing Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

D2Direction Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : D2Direction Property (ICurveDrivenPatternFeatureData)

Gets or sets Direction 2 of this bidirectional curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2Direction As System.Object

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Object  
  
instance.D2Direction = value  
value = instance.D2Direction
```

C#

```
System.object D2Direction {get; set;}
```

C++/CLI

```
property System.Object^ D2Direction {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pattern direction entity

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::D2Direction](#).

Remarks

The pattern direction can be a curve, edge, sketch entity, or a sketch. You must pre-select the Direction 2 entity using selection Mark = 2 before creating the feature definition. Use this property only when editing the pattern feature.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::D1Direction Property](#)
[ICurveDrivenPatternFeatureData::Dir2Specified Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

D2InstanceCount Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : D2InstanceCount Property (ICurveDrivenPatternFeatureData)

Gets or sets the number of instances in Direction 2 of this bidirectional curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2InstanceCount As System.Integer

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Integer
```

```
instance.D2InstanceCount = value
```

```
value = instance.D2InstanceCount
```

C#

```
System.int D2InstanceCount {get; set;}
```

C++/CLI

```
property System.int D2InstanceCount {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Number of instances

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::D2InstanceCount](#).

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::D1InstanceCount Property](#)
[ICurveDrivenPatternFeatureData::Dir2Specified Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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D2IsEqualSpaced Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : D2IsEqualSpaced Property (ICurveDrivenPatternFeatureData)

Gets or sets whether the pattern items are equally spaced in Direction 2 of this bidirectional curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property D2IsEqualSpaced As System.Boolean

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Boolean  
  
instance.D2IsEqualSpaced = value  
value = instance.D2IsEqualSpaced
```

C#

```
System.bool D2IsEqualSpaced {get; set;}
```

C++/CLI

```
property System.bool D2IsEqualSpaced {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True spaces the pattern items equally, false does not

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::D2IsEqualSpaced](#).

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::D1IsEqualSpaced Property](#)
[ICurveDrivenPatternFeatureData::Dir2Specified Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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D2PatternSeedOnly Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : D2PatternSeedOnly Property (ICurveDrivenPatternFeatureData)

Gets or sets whether to replicate the seed pattern only in Direction 2, without replicating the curve pattern created under Direction 1.

.NET Syntax

Visual Basic (Declaration)

Property D2PatternSeedOnly As System.Boolean

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Boolean  
  
instance.D2PatternSeedOnly = value  
value = instance.D2PatternSeedOnly
```

C#

```
System.bool D2PatternSeedOnly {get; set;}
```

C++/CLI

```
property System.bool D2PatternSeedOnly {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True patterns the seed feature only in Direction 2

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::D2PatternSeedOnly](#).

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::Dir2Specified Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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D2ReverseDirection Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : D2ReverseDirection Property (ICurveDrivenPatternFeatureData)

Gets or sets whether to reverse Direction 2 for this bidirectional curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property D2ReverseDirection As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Boolean
```

```
instance.D2ReverseDirection = value
```

```
value = instance.D2ReverseDirection
```

C#

```
System.bool D2ReverseDirection {get; set;}
```

C++/CLI

```
property System.bool D2ReverseDirection {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True reverses Direction 2, false does not

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::D2ReverseDirection](#).

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::D1ReverseDirection Property](#)
[ICurveDrivenPatternFeatureData::Dir2Specified Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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D2Spacing Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : D2Spacing Property (ICurveDrivenPatternFeatureData)

Gets or sets the spacing of pattern instances in Direction 2 for this bidirectional curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Property D2Spacing As System.Double
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Double  
  
instance.D2Spacing = value  
value = instance.D2Spacing
```

C#

```
System.double D2Spacing {get; set;}
```

C++/CLI

```
property System.double D2Spacing {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pattern spacing

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::D2Spacing](#).

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::D1Spacing Property](#)
[ICurveDrivenPatternFeatureData::Dir2Specified Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

Dir2Specified Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : Dir2Specified Property (ICurveDrivenPatternFeatureData)

Gets or sets whether Direction 2 is specified for this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property Dir2Specified As System.Boolean

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Boolean
```

```
instance.Dir2Specified = value
```

```
value = instance.Dir2Specified
```

C#

```
System.bool Dir2Specified {get; set;}
```

C++/CLI

```
property System.bool Dir2Specified {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if Direction 2 is specified, false if not

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::Dir2Specified](#).

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::D2Direction Property](#)
[ICurveDrivenPatternFeatureData::D2InstanceCount Property](#)
[ICurveDrivenPatternFeatureData::D2IsEqualSpaced Property](#)
[ICurveDrivenPatternFeatureData::D2PatternSeedOnly Property](#)
[ICurveDrivenPatternFeatureData::D2ReverseDirection Property](#)
[ICurveDrivenPatternFeatureData::D2Spacing Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

FeatureScope Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : FeatureScope Property (ICurveDrivenPatternFeatureData)

Gets which bodies in this multibody part are affected by this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property FeatureScope As System.Boolean

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Boolean
```

```
value = instance.FeatureScope
```

C#

```
System.bool FeatureScope {get;}
```

C++/CLI

```
property System.bool FeatureScope {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to specify affected bodies, false to apply the pattern to all bodies every time the feature regenerates

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::FeatureScope](#).

Remarks

This property is valid only if [ICurveDrivenPatternFeatureData::GeometryPattern](#) is true.

Call [ICurveDrivenPatternFeatureData::SetFeatureScope](#) to set this property.

For more information, see the **Curve Driven Patterns and the Curve Driven Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FeatureScopeBodies Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : FeatureScopeBodies Property (ICurveDrivenPatternFeatureData)

Gets the bodies in this multibody part that are affected by this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property FeatureScopeBodies As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Object
```

```
value = instance.FeatureScopeBodies
```

C#

```
System.Object FeatureScopeBodies {get;}
```

C++/CLI

```
property System.Object^ FeatureScopeBodies {  
    System.Object^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [bodies](#) to be affected

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::FeatureScopeBodies](#).

Remarks

This property is valid only if [ICurveDrivenPatternFeatureData::GeometryPattern](#) is true.

Call [ICurveDrivenPatternFeatureData::SetFeatureScope](#) to set this property.

For more information, see the **Curve Driven Patterns and the Curve Driven Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GeometryPattern Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : GeometryPattern Property (ICurveDrivenPatternFeatureData)

Gets or sets whether to create the pattern using only the geometry (faces and edges) of the feature.

.NET Syntax

Visual Basic (Declaration)

```
Property GeometryPattern As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Boolean
```

```
instance.GeometryPattern = value
```

```
value = instance.GeometryPattern
```

C#

```
System.bool GeometryPattern {get; set;}
```

C++/CLI

```
property System.bool GeometryPattern {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use only the geometry of the feature to create the pattern (faster), false to pattern and solve each instance of the feature in its entirety (slower)

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::GeometryPattern](#).

Remarks

This property is valid only if [ICurveDrivenPatternFeatureData::BodyPattern](#) is false.

If this property is true, you can call [ICurveDrivenPatternFeatureData::SetFeatureScope](#).

For more information, see the **Curve Driven Patterns and the Curve Driven Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPatternBodyCount Method (ICurveDrivenPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : GetPatternBodyCount Method (ICurveDrivenPatternFeatureData)

Gets the number of seed bodies in this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPatternBodyCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.GetPatternBodyCount()
```

C#

```
System.int GetPatternBodyCount()
```

C++/CLI

```
System.int GetPatternBodyCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of seed bodies

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::GetPatternBodyCount](#).

Example

[Create and Access Curve-driven Pattern Feature \(C#\)](#)
[Create and Access Curve-driven Pattern Feature \(VB.NET\)](#)
[Create and Access Curve-driven Pattern Feature \(VBA\)](#)

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::IGetPatternBodyArray Method](#)
[ICurveDrivenPatternFeatureData::ISetPatternBodyArray Method](#)
[ICurveDrivenPatternFeatureData::PatternBodyArray Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPatternFaceCount Method (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : GetPatternFaceCount Method (ICurveDrivenPatternFeatureData)

Gets a list of pattern faces for this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPatternFaceCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Integer  
  
value = instance.GetPatternFaceCount()
```

C#

```
System.int GetPatternFaceCount()
```

C++/CLI

```
System.int GetPatternFaceCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of faces

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::GetPatternFaceCount](#).

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::IGetPatternFaceArray Method](#)
[ICurveDrivenPatternFeatureData::ISetPatternFaceArray Method](#)
[ICurveDrivenPatternFeatureData::PatternFaceArray Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetPatternFeatureCount Method (ICurveDrivenPatternFeatureData)

[See Also](#) [Example](#)



Gets the number of seed features for this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPatternFeatureCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.GetPatternFeatureCount()
```

C#

```
System.int GetPatternFeatureCount()
```

C++/CLI

```
System.int GetPatternFeatureCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of features

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::GetPatternFeatureCount](#).

Example

[Create and Access Curve-driven Pattern Feature \(C#\)](#)
[Create and Access Curve-driven Pattern Feature \(VB.NET\)](#)
[Create and Access Curve-driven Pattern Feature \(VBA\)](#)

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::IGetPatternFeatureArray Method](#)
[ICurveDrivenPatternFeatureData::ISetPatternFeatureArray Method](#)
[ICurveDrivenPatternFeatureData::PatternFeatureArray Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSkippedItemCount Method (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : GetSkippedItemCount Method (ICurveDrivenPatternFeatureData)

Gets the number of skipped items for this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSkippedItemCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.GetSkippedItemCount()
```

C#

```
System.int GetSkippedItemCount()
```

C++/CLI

```
System.int GetSkippedItemCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of skipped items

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::GetSkippedItemCount](#).

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::IGetSkippedItemArray Method](#)
[ICurveDrivenPatternFeatureData::ISetSkippedItemArray Method](#)
[ICurveDrivenPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2001Plus FCS, revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTransform Method (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : GetTransform Method (ICurveDrivenPatternFeatureData)

Gets the transform for the specified instance of this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTransform( _  
    ByVal Instance As System.Integer _  
) As MathTransform
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim Instance As System.Integer  
Dim value As MathTransform
```

```
value = instance.GetTransform(Instance)
```

C#

```
MathTransform GetTransform(  
    System.int Instance  
)
```

C++/CLI

```
MathTransform^ GetTransform(  
    & System.int Instance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Instance
Index of one repeating element in this pattern (see **Remarks**)

Return Value

[Transform](#)

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::GetTransform](#).

Remarks

1 <= *Instance* <= ([ICurveDrivenPatternFeatureData::D1InstanceCount](#) + [ICurveDrivenPatternFeatureData::D2InstanceCount](#))

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IAccessSelections Method (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : IAccessSelections Method (ICurveDrivenPatternFeatureData)

Gains access to the selections that define this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IAccessSelections( _  
    ByVal TopDoc As ModelDoc2, _  
    ByVal Component As Component2 _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim TopDoc As ModelDoc2  
Dim Component As Component2  
Dim value As System.Boolean  
  
value = instance.IAccessSelections(TopDoc, Component)
```

C#

```
System.bool IAccessSelections(  
    ModelDoc2 TopDoc,  
    Component2 Component  
)
```

C++/CLI

```
System.bool IAccessSelections(  
& ModelDoc2^ TopDoc,  
& Component2^ Component  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc
Top-level document

Component
Component in which the feature is to be modified

Return Value

True if the selections are successfully accessed, false if not

 Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::IAccessSelections](#).

 Remarks

Remark

To modify a feature in a...	Then...
Part	• TopDoc argument is the IModelDoc2 for the part • Component argument is NULL
Assembly	• TopDoc is the IModelDoc2 object for the assembly • Component argument is the IComponent2 object in which the feature is to be modified

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) if you modified the feature
- [ICurveDrivenPatternFeatureData::ReleaseSelectionAccess](#) if you did not

 See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::AccessSelections Method](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetPatternBodyArray Method (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : IGetPatternBodyArray Method (ICurveDrivenPatternFeatureData)

Gets a list of the seed bodies for this curve-driven pattern feature.

 .NET Syntax

Visual Basic (Declaration)

Function IGetPatternBodyArray() As Body2

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As Body2  
  
value = instance.IGetPatternBodyArray()
```

C#

Body2 IGetPatternBodyArray()

C++/CLI

Body2^ IGetPatternBodyArray();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of seed [bodies](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::GetPatternBodyCount Method](#)
[ICurveDrivenPatternFeatureData::ISetPatternBodyArray Method](#)
[ICurveDrivenPatternFeatureData::PatternBodyArray Property](#)
[ICurveDrivenPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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IGetPatternFaceArray Method (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : IGetPatternFaceArray Method (ICurveDrivenPatternFeatureData)

Gets a list of pattern faces for this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPatternFaceArray() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Object  
  
value = instance.IGetPatternFaceArray()
```

C#

```
System.object IGetPatternFaceArray()
```

C++/CLI

```
System.Object^ IGetPatternFaceArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of pattern [faces](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::GetPatternFaceCount Method](#)
[ICurveDrivenPatternFeatureData::ISetPatternFaceArray Method](#)
[ICurveDrivenPatternFeatureData::PatternFaceArray Property](#)
[ICurveDrivenPatternFeatureData::BodyPattern Property](#)

Availability

IGetPatternFeatureArray Method (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : IGetPatternFeatureArray Method (ICurveDrivenPatternFeatureData)

Gets a list of pattern features in this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPatternFeatureArray() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Object
```

```
value = instance.IGetPatternFeatureArray()
```

C#

```
System.object IGetPatternFeatureArray()
```

C++/CLI

```
System.Object^ IGetPatternFeatureArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of pattern [features](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::ISetPatternFeatureArray Method](#)
[ICurveDrivenPatternFeatureData::GetPatternFeatureCount Method](#)
[ICurveDrivenPatternFeatureData::IGetPatternFeatureArray Property](#)
[ICurveDrivenPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

IGetSkippedItemArray Method (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : IGetSkippedItemArray Method (ICurveDrivenPatternFeatureData)

Gets the array of integers representing the skipped items for this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSkippedItemArray() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData
Dim value As System.Integer

value = instance.IGetSkippedItemArray()
```

C#

```
System.int IGetSkippedItemArray()
```

C++/CLI

```
System.int IGetSkippedItemArray();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of integers representing skipped items
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

This is a 0-based array, so if you skip the third and fifth items in the pattern, then:

- Array[0] = 3
- Array[1] = 5

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::GetSkippedItemCount Method](#)
[ICurveDrivenPatternFeatureData::ISetSkippedItemArray Method](#)
[ICurveDrivenPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetPatternBodyArray Method (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : ISetPatternBodyArray Method (ICurveDrivenPatternFeatureData)

Sets the list of seed bodies for this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetPatternBodyArray( _
    ByVal BodyCount As System.Integer, _
    ByRef ArrayDataIn As Body2 _
)
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData
Dim BodyCount As System.Integer
Dim ArrayDataIn As Body2

instance.ISetPatternBodyArray(BodyCount, ArrayDataIn)
```

C#

```
void ISetPatternBodyArray(
    System.int BodyCount,
```

```
ref Body2 ArrayDataIn
)
```

C++/CLI

```
void ISetPatternBodyArray(
& System.int BodyCount,
& Body2^% ArrayDataIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BodyCount

Number of seed bodies

ArrayDataIn

- in-process, unmanaged C++: Pointer to an array of seed [bodies](#) of size BodyCount
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::GetPatternBodyCount Method](#)
[ICurveDrivenPatternFeatureData::IGetPatternBodyArray Method](#)
[ICurveDrivenPatternFeatureData::IPatternBodyArray Property](#)
[ICurveDrivenPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetPatternFaceArray Method (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : ISetPatternFaceArray Method (ICurveDrivenPatternFeatureData)

Sets a list of pattern faces for this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetPatternFaceArray( _
    ByVal FaceCount As System.Integer, _
    ByRef ArrayDataIn As System.Object _
)
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData
Dim FaceCount As System.Integer
Dim ArrayDataIn As System.Object
```

```
instance.ISetPatternFaceArray(FaceCount, ArrayDataIn)
```

C#

```
void ISetPatternFaceArray(
    System.int FaceCount,
    ref System.object ArrayDataIn
)
```

C++/CLI

```
void ISetPatternFaceArray(  
& System.int FaceCount,  
& System.Object^% ArrayDataIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FaceCount

Number of pattern faces

ArrayDataIn

- in-process, unmanaged C++: Pointer to an array of pattern faces of size FaceCount

• VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICurveDrivenPatternFeatureData Interface](#)

[ICurveDrivenPatternFeatureData Members](#)

[ICurveDrivenPatternFeatureData::GetPatternFaceCount Method](#)

[ICurveDrivenPatternFeatureData::GetPatternFaceArray Method](#)

[ICurveDrivenPatternFeatureData::PatternFaceArray Property](#)

[ICurveDrivenPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ISetPatternFeatureArray Method (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : ISetPatternFeatureArray Method (ICurveDrivenPatternFeatureData)

Sets the list of pattern features for this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetPatternFeatureArray( _  
    ByVal FeatCount As System.Integer, _  
    ByRef ArrayDataIn As System.Object _  
)
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim FeatCount As System.Integer  
Dim ArrayDataIn As System.Object
```

```
instance.ISetPatternFeatureArray(FeatCount, ArrayDataIn)
```

C#

```
void ISetPatternFeatureArray(  
    System.int FeatCount,  
    ref System.object ArrayDataIn  
)
```

C++/CLI

```
void ISetPatternFeatureArray(  
& System.int FeatCount,
```

```
& System.Object^% ArrayDataIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatCount

Number of pattern features

ArrayDataIn

- in-process, unmanaged C++: Pointer to an array of pattern features of size FeatCount

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICurveDrivenPatternFeatureData Interface](#)

[ICurveDrivenPatternFeatureData Members](#)

[ICurveDrivenPatternFeatureData::GetPatternFeatureCount Method](#)

[ICurveDrivenPatternFeatureData::GetPatternFeatureArray Method](#)

[ICurveDrivenPatternFeatureData::PatternFeatureArray Property](#)

[ICurveDrivenPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ISetSkippedItemArray Method (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : ISetSkippedItemArray Method (ICurveDrivenPatternFeatureData)

Sets the list of skipped items for this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetSkippedItemArray( _
    ByVal FeatCount As System.Integer, _
    ByRef ArrayDataIn As System.Integer _
)
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData
Dim FeatCount As System.Integer
Dim ArrayDataIn As System.Integer
```

```
instance.ISetSkippedItemArray(FeatCount, ArrayDataIn)
```

C#

```
void ISetSkippedItemArray(
    System.int FeatCount,
    ref System.int ArrayDataIn
)
```

C++/CLI

```
void ISetSkippedItemArray(
    & System.int FeatCount,
    & System.int% ArrayDataIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatCount

Number of skipped features

ArrayDataIn

- in-process, unmanaged C++: Pointer to an array of skipped items of size FeatCount
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

See [Accessing Selections that Define Features](#) for additional details on using this method.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::GetSkippedItemCount Method](#)
[ICurveDrivenPatternFeatureData::GetSkippedItemArray Method](#)
[ICurveDrivenPatternFeatureData::SkippedItemArray Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PatternBodyArray Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : PatternBodyArray Property (ICurveDrivenPatternFeatureData)

Gets or sets a list of the bodies to pattern for this curve-driven pattern.

.NET Syntax

Visual Basic (Declaration)

Property PatternBodyArray As System.Object

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Object
```

```
instance.PatternBodyArray = value
```

```
value = instance.PatternBodyArray
```

C#

```
System.Object PatternBodyArray {get; set;}
```

C++/CLI

```
property System.Object^ PatternBodyArray {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [bodies](#) to pattern

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::PatternBodyArray](#).

Remarks

This property is valid only if:

- you have a multibody part,
- and -
- [ICurveDrivenPatternFeatureData::BodyPattern](#) is true.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::GetPatternBodyCount Method](#)
[ICurveDrivenPatternFeatureData::IGetPatternBodyArray Method](#)
[ICurveDrivenPatternFeatureData::ISetPatternBodyArray Method](#)
[ICurveDrivenPatternFeatureData::BodyPattern Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PatternElement Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : PatternElement Property (ICurveDrivenPatternFeatureData)

Gets or sets the type of entities to base this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property PatternElement As System.Integer

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Integer
```

```
instance.PatternElement = value
```

```
value = instance.PatternElement
```

C#

```
System.int PatternElement {get; set;}
```

C++/CLI

```
property System.int PatternElement {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of entities to base this curve-driven pattern feature as defined in [swPatternElementSelection_e](#)

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::PatternElement](#).

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::BodyPattern Property\(\)](#)

Availability

PatternFaceArray Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : PatternFaceArray Property (ICurveDrivenPatternFeatureData)

Gets or sets the list of faces to include in this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property PatternFaceArray As System.Object

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData
Dim value As System.Object

instance.PatternFaceArray = value
value = instance.PatternFaceArray
```

C#

```
System.Object PatternFaceArray {get; set;}
```

C++/CLI

```
property System.Object^ PatternFaceArray {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of pattern [faces](#)

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::PatternFaceArray](#).

Remarks

This property is valid only if [ICurveDrivenPatternFeatureData::BodyPattern](#) is false.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::GetPatternFaceCount Method](#)
[ICurveDrivenPatternFeatureData::IGetPatternFaceArray Method](#)
[ICurveDrivenPatternFeatureData::ISetPatternFaceArray Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

PatternFeatureArray Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : PatternFeatureArray Property (ICurveDrivenPatternFeatureData)

Gets or sets the list of features to include in this curve-driven feature pattern.

Visual Basic Syntax

Visual Basic (Declaration)

Property PatternFeatureArray As System.Object

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Object  
  
instance.PatternFeatureArray = value  
value = instance.PatternFeatureArray
```

C#

```
System.object PatternFeatureArray {get; set;}
```

C++/CLI

```
property System.Object^ PatternFeatureArray {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of [features](#)

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::PatternFeatureArray](#).

Remarks

This property is valid only if [ICurveDrivenPatternFeatureData::BodyPattern](#) is false.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::GetPatternFeatureCount Method](#)
[ICurveDrivenPatternFeatureData::IGetPatternFeatureArray Method](#)
[ICurveDrivenPatternFeatureData::ISetPatternFeatureArray Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PropagateVisualProperty Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : PropagateVisualProperty Property (ICurveDrivenPatternFeatureData)

Gets or sets whether to propagate visual properties (e.g., colors, textures, etc.) to all curve-driven pattern instances.

Visual Basic Syntax

Visual Basic (Declaration)

Property PropagateVisualProperty As System.Boolean

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData
Dim value As System.Boolean

instance.PropagateVisualProperty = value
value = instance.PropagateVisualProperty
```

C#

```
System.bool PropagateVisualProperty {get; set;}
```

C++/CLI

```
property System.bool PropagateVisualProperty {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to propagate all visual properties, false to not

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::PropagateVisualProperty](#).

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReleaseSelectionAccess Method (ICurveDrivenPatternFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : ReleaseSelectionAccess Method (ICurveDrivenPatternFeatureData)

Releases access to the selections that define this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ReleaseSelectionAccess()
```

Visual Basic (Usage)

```
Dim instance As ICurveDrivenPatternFeatureData

instance.ReleaseSelectionAccess()
```

C#

```
void ReleaseSelectionAccess()
```

C++/CLI

```
void ReleaseSelectionAccess();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::ReleaseSelectionAccess](#).

Example

[Create and Access Curve-driven Pattern Feature \(C#\)](#)
[Create and Access Curve-driven Pattern Feature \(VB.NET\)](#)
[Create and Access Curve-driven Pattern Feature \(VBA\)](#)

Remarks

[ICurveDrivenPatternFeatureData::AccessSelections](#) puts the model into a rollback state to allow access to the selections that define this feature. Use this method to restore the rollback state if you did not modify the feature or use [IFeature::ModifyDefinition](#) if you did.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetFeatureScope Method (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : SetFeatureScope Method (ICurveDrivenPatternFeatureData)

Sets the feature scope, whether to autoselect the affected bodies, and the affected bodies in this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

```
Function SetFeatureScope( _  
    ByVal FeatureScopeOption As System.Boolean, _  
    ByVal AutoSelectBodies As System.Boolean, _  
    ByVal Bodies As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim FeatureScopeOption As System.Boolean  
Dim AutoSelectBodies As System.Boolean  
Dim Bodies As System.Object  
Dim value As System.Boolean  
  
value = instance.SetFeatureScope(FeatureScopeOption, AutoSelectBodies, Bodies)
```

C#

```
System.bool SetFeatureScope(  
    System.bool FeatureScopeOption,  
    System.bool AutoSelectBodies,  
    System.object Bodies  
)
```

C++/CLI

```
System.bool SetFeatureScope(  
& System.bool FeatureScopeOption,  
& System.bool AutoSelectBodies,  
& System.Object^ Bodies  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatureScopeOption

True to specify affected bodies, false to apply the pattern to all bodies every time the feature regenerates (see **Remarks**)

AutoSelectBodies

True to automatically select all bodies intersected by this pattern feature, false to specify affected bodies (see **Remarks**)

Bodies

Array of [bodies](#) to be affected; valid only if FeatureScopeOption is true and AutoSelectBodies is false

Return Value

True if feature scope set successfully, false if not (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::SetFeatureScope](#).

Remarks

If this method returns false, then default values for FeatureScopeOption and AutoSelectBodies are set. A subsequent call to [IFeatureManager::CreateFeature](#) creates the feature with a FeatureScopeOption of true and an AutoSelectBodies of true.

After calling IFeature::ModifyDefinition, call [IFeature::GetErrorCode2](#) to determine what's wrong and then take necessary remedial action.

For more information, see the **Curve Driven Patterns and the Curve Driven Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See Also

[ICurveDrivenPatternFeatureData Interface](#)

[ICurveDrivenPatternFeatureData Members](#)

[ICurveDrivenPatternFeatureData::AutoSelect Property \(\)](#)

[ICurveDrivenPatternFeatureData::FeatureScope Property \(\)](#)

[ICurveDrivenPatternFeatureData::FeatureScopeBodies Property \(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SkippedItemArray Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : SkippedItemArray Property (ICurveDrivenPatternFeatureData)

Gets or sets the skipped items for this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property SkippedItemArray As System.Object

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Object
```

```
instance.SkippedItemArray = value
```

```
value = instance.SkippedItemArray
```

C#

```
System.object SkippedItemArray {get; set;}
```

C++/CLI

```
property System.Object^ SkippedItemArray {  
    System.Object^ get();  
    void set ( &    System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of integers representing the skipped items

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::SkippedItemArray](#).

Remarks

This array is 0-based. If you skip the third and fifth items, then the array looks like ArrayOut[0]=3 and ArrayOut[1]=5.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)
[ICurveDrivenPatternFeatureData::GetSkippedItemCount Method](#)
[ICurveDrivenPatternFeatureData::IGetSkippedItemArray Method](#)
[ICurveDrivenPatternFeatureData::ISetSkippedItemArray Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

VarySketch Property (ICurveDrivenPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICurveDrivenPatternFeatureData Interface](#) : VarySketch Property (ICurveDrivenPatternFeatureData)

Gets or sets whether to allow pattern instances of a seed sketch to vary in relation to enclosing geometry in this curve-driven pattern feature.

.NET Syntax

Visual Basic (Declaration)

Property VarySketch As System.Boolean

Visual Basic (Usage)



```
Dim instance As ICurveDrivenPatternFeatureData  
Dim value As System.Boolean
```

```
instance.VarySketch = value
```

```
value = instance.VarySketch
```

C#

```
System.bool VarySketch {get; set;}
```

C++/CLI

```
property System.bool VarySketch {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to allow pattern instances of a seed sketch to vary in relation to enclosing geometry, false to not

Visual Basic for Applications (VBA) Syntax

See [CurveDrivenPatternFeatureData::VarySketch](#).

Remarks

This property is valid only when creating a pattern from a seed sketch.

The seed sketch has:

- An enclosing boundary of base geometry.
- Fully defined relations.
- Dimensions only for measurements that will not vary in the pattern instances.
- A driving dimension that is used to specify the direction of the curve-driven pattern.

If this property is true:

- Specify [ICurveDrivenPatternFeatureData.D1Direction](#) with a driving dimension of the seed sketch.
- Pattern instances vary with and are constrained by the enclosing geometry.

See the **SOLIDWORKS Help > Parts and Features > Features > Patterns and Mirroring > Control and Modify Patterns > Vary Sketch** topic.

See [Accessing Selections that Define Features](#) for additional details on using this property.

See Also

[ICurveDrivenPatternFeatureData Interface](#)
[ICurveDrivenPatternFeatureData Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0